

Like Negi, countless others are discovering what stress is doing to their bodies. As Dr B.N. Gangadhar, former head of NIMHANS (National Institute of Mental Health and Neurosciences), Bengaluru, points out, "Stress is no longer just a mental condition. It is also linked to a vast number of physical problems." Scientific research in the past two years has alerted us to the fact that stress is contributing directly or indirectly to heart disease, brain stroke, diabetes, cancer, lung ailments, liver cirrhosis, infertility, besides addiction, obesity and depression leading to suicide, and is one of the leading causes of death. Yet people do not recognise stress as a problem until it manifests in more deadly disease. As Dr Sunil Kumar Mishra, director of the endocrinology and diabetology practice at the Medanta hospital in Gurugram, says, "People usually don't take something seriously till it is presented to them in black and white. Since you can't really measure stress in your body, it is only when a major episode takes place that you realise the impact of your overstressed lifestyle."

And the Covid-19 pandemic—both the virus itself, and the measures to deal with it—has only increased the impact of already stressful lives, exacting a heavy toll on the body. "There is an uptick in sudden heart attacks, for example," says Dr Ashok Seth, chairman, Fortis Escorts Heart Institute, "with stress being a significant contributing factor." The curtains have scarcely come down on the Covid pandemic and India is having to acknowledge the dangers of its other epidemic, a silent one, brought on by stress. In a December 2020 survey of 10,000 Indians conducted by The Centre of Healing (TCOH), a mental health service platform based in Delhi, 74 per cent of the respondents reported stress, and 88 per cent anxiety. Nearly 70 per cent of the therapists surveyed documented an increase in the number of patient visits; 55 per cent said there had been an increase in first-time clients after the pandemic.

Stress is no stranger to humankind; as a normal bodily function, it is as old as existence itself. The human body is primed both to recognise it, and to deal with it, through a combination of responses that make up what is called the "fight or flight" mechanism. It is the over-reaction to stress where things start to go wrong, and managing it becomes imperative, for one's overall physical and mental well-being. Scientific energies in the past few years, therefore, have concentrated on understanding the body's response to stress, at what stage it becomes a problem and how one can rein it in to regain control of one's physical and psychological life. However, to assess what havoc prolonged stress can wreak on the human body in the long term, one has to understand the science behind it first.

> "Chronic stress can disturb **dopamine, serotonin and reproductive hormones**, which could cause depression, low libido, infertility, PCOS"
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> **–DR SAMEER MALHOTRA**
> *Director, Mental Health Department, Max Super Speciality Hospital, Delhi*



THE BIOLOGY OF STRESS

The key to the body's response to stress lies in its endocrine system, along what is called the hypothalamic-pituitary-adrenal (HPA) axis. The hypothalamus is the command centre of the brain, regulating a multitude of things such as mood, the body temperature, heart rate, food intake, sex drive, energy, thirst and the sleep cycle. It also controls the secretion of hormones—chemicals that travel along the bloodstream to various tissues and organs in the body, activating and coordinating several functions.

When the brain senses a stressor—be it physical danger, the boss yelling at you, a piece of bad news, or just everyday stress—it sets off a series of well-orchestrated physiological processes. The hypothalamus releases the corticotrophin-releasing hormone (CRH), which carries the message to the control room of the endocrine system—the pea-sized pituitary gland located just below it. As though in a relay race, the pituitary gland takes the baton and sends another message down the bloodstream via the adrenocorticotropic